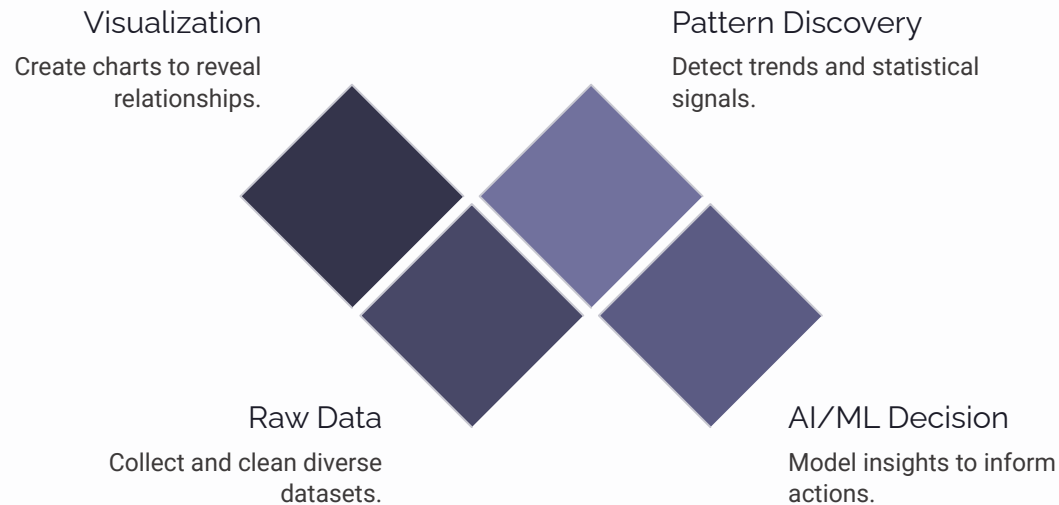


Data Visualization & Exploratory Data Analysis

Module 2 — Probability & Statistics · Lesson 3

"Turning Numbers Into Insights"



Can You Tell Me What's Happening in This Data?

Look at this table. You have 20 seconds. What is the trend? What is the pattern? Who is performing well?

Student	Marks	Student	Marks	Student	Marks
A	42	G	71	M	58
B	48	H	91	N	66
C	55	I	37	O	79
D	63	J	84	P	43
E	68	K	52	Q	88
F	75	L	60	R	96

📄 🤯 Too many numbers. Patterns are hidden. Your brain is working too hard. **What if you could simply SEE the pattern?**

Why Visualization? Because Seeing Is Understanding

A dataset *contains* information. A visualization *makes* that information visible. The moment you plot data, patterns that were invisible in a table suddenly jump out at you.



Trends

Is something increasing or decreasing over time?



Spread

Are values tightly packed or widely scattered?



Outliers

Is there one value that stands far apart from the crowd?



Relationships

Do two variables move together?

"Numbers are the ingredients. Visualization is the dish that lets you understand what you're working with."

Real-World Analogy: Map vs. List of Locations



A Plain List

Delhi · Mumbai · Patna · Kolkata ·
Bengaluru · Chennai · Jaipur · Hyderabad

All the information is there. But can you tell which cities are close? Which are far apart? What region they belong to?



A Map

The same cities on a map instantly give you **structure, distance, and spatial relationships**.

A list gives you *data*. A map gives you *understanding*.

→ This is exactly what a graph does for your numbers.



Key Insight: Raw data gives information. Visualization gives structure. In AI/ML, structure is everything.

The Big 3 Questions of Data Visualization

Before picking any graph, ask yourself one of three simple questions. These three questions cover 90% of exploratory analysis in real AI/ML projects.



What does the data look like?

How are values distributed? Are they
bunched together or spread out? → **Use a
Histogram**



Are there unusual values?

Is there a value that seems too high or too
low? → **Use a Box Plot**

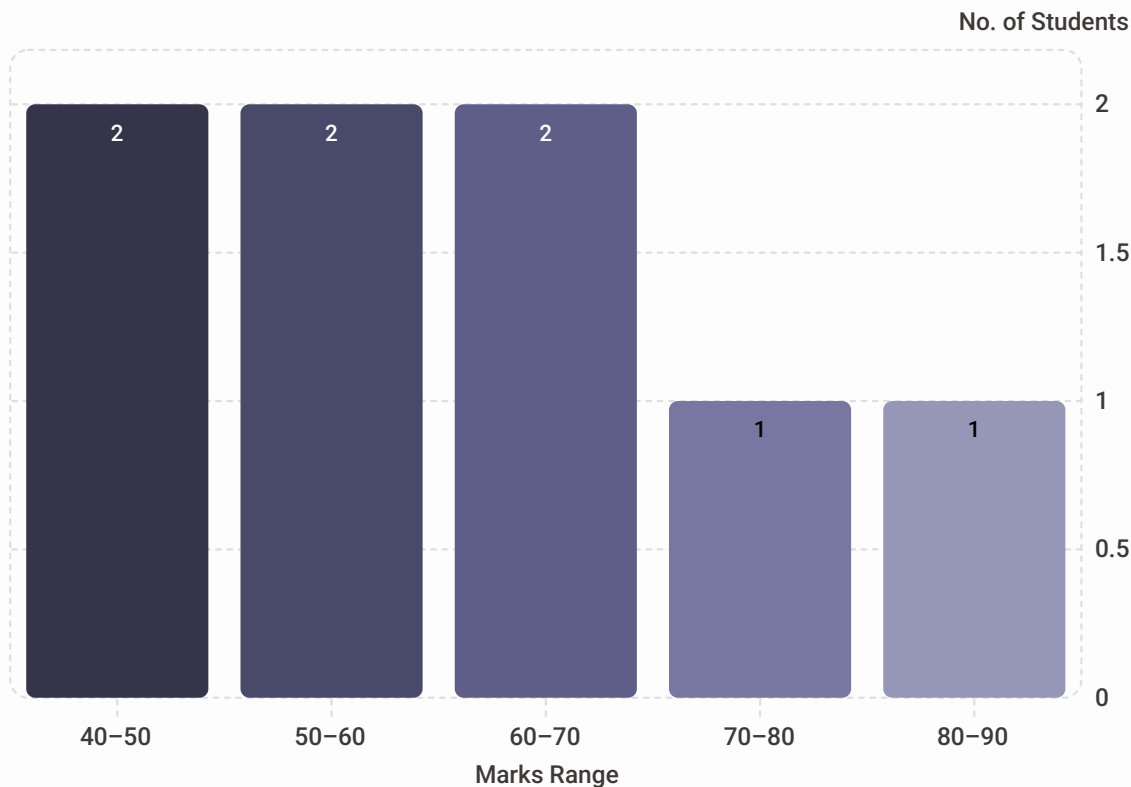


Are two variables related?

When one variable changes, does another
also change? → **Use a Scatter Plot**

This three-question framework is your mental model for the entire lesson. Memorise it.

Histogram — "How Are the Values Distributed?"



How to Read It

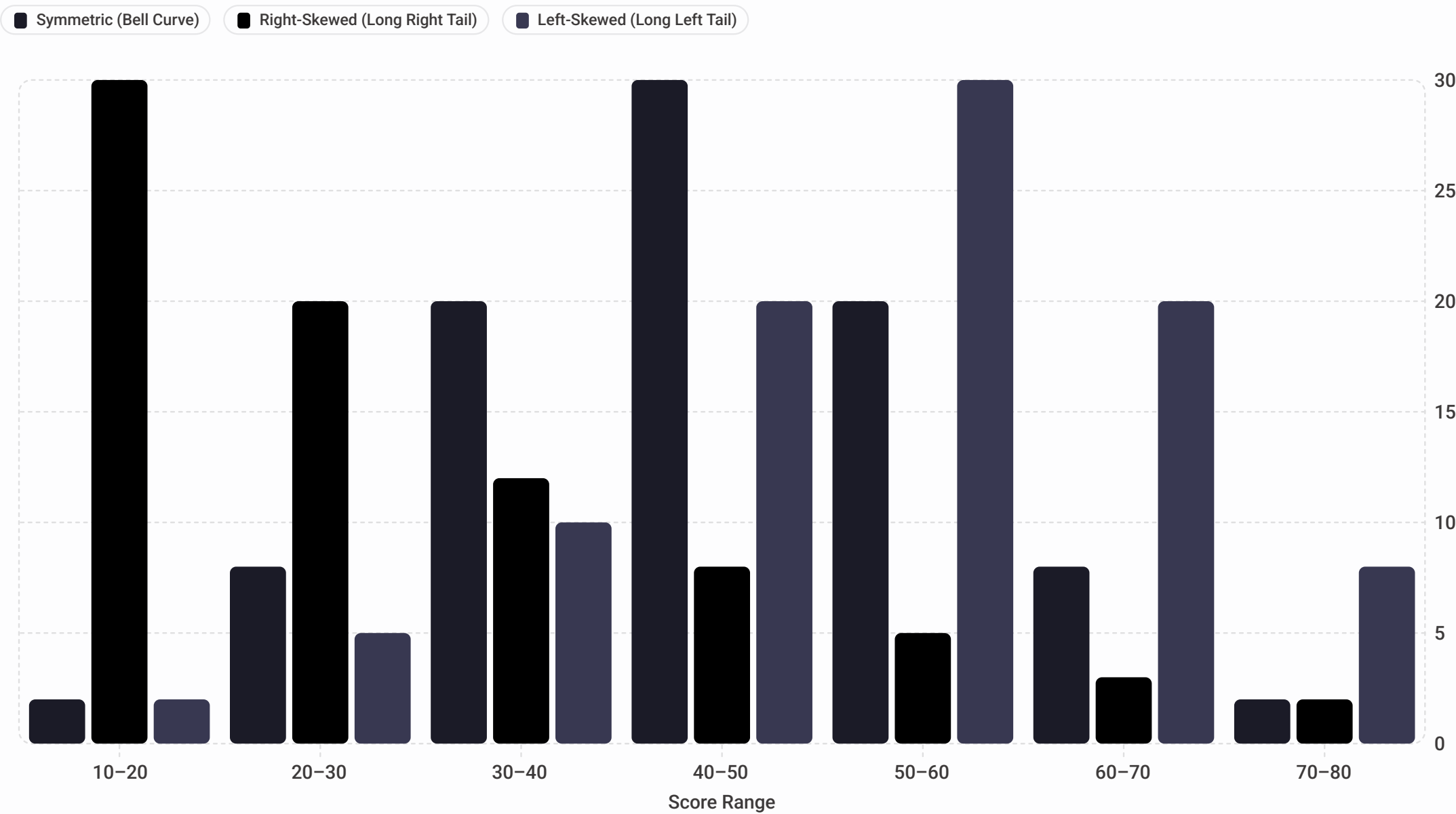
- **X-axis** → Value ranges (called "bins")
- **Y-axis** → Count of observations in that bin
- **Bar height** → How many students scored in that range
- **Bins** → Group values into intervals for clarity



💡 **Analogy:** A histogram is like counting how many students sit in each row of a classroom, sorted by their marks.

Understanding Distribution Shape — The Tail Tells the Story

The *shape* of a histogram tells you a great deal about your data — before you calculate a single number. Look at the tail: the direction it stretches tells you the direction of skew.



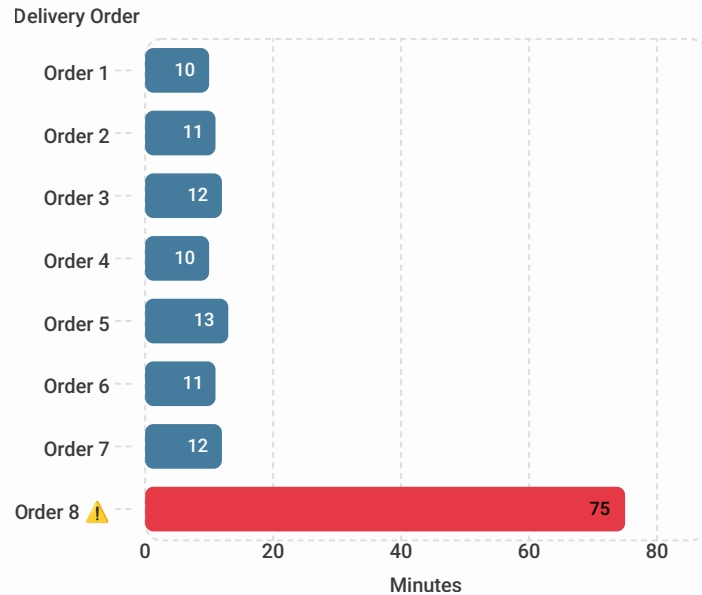
Symmetric
Equal on both sides. Mean $\approx$ Median.
Common in natural measurements.

Right-Skewed
Long tail on the right. A few very high values pull the mean up. Common in incomes.

Left-Skewed
Long tail on the left. A few very low values pull the mean down. Exam scores near full marks.

Outliers — The Unusual Data Point

Consider average food delivery times (in minutes): 10, 11, 12, 10, 13, 11, 12, **75**



🔍 What should we investigate?

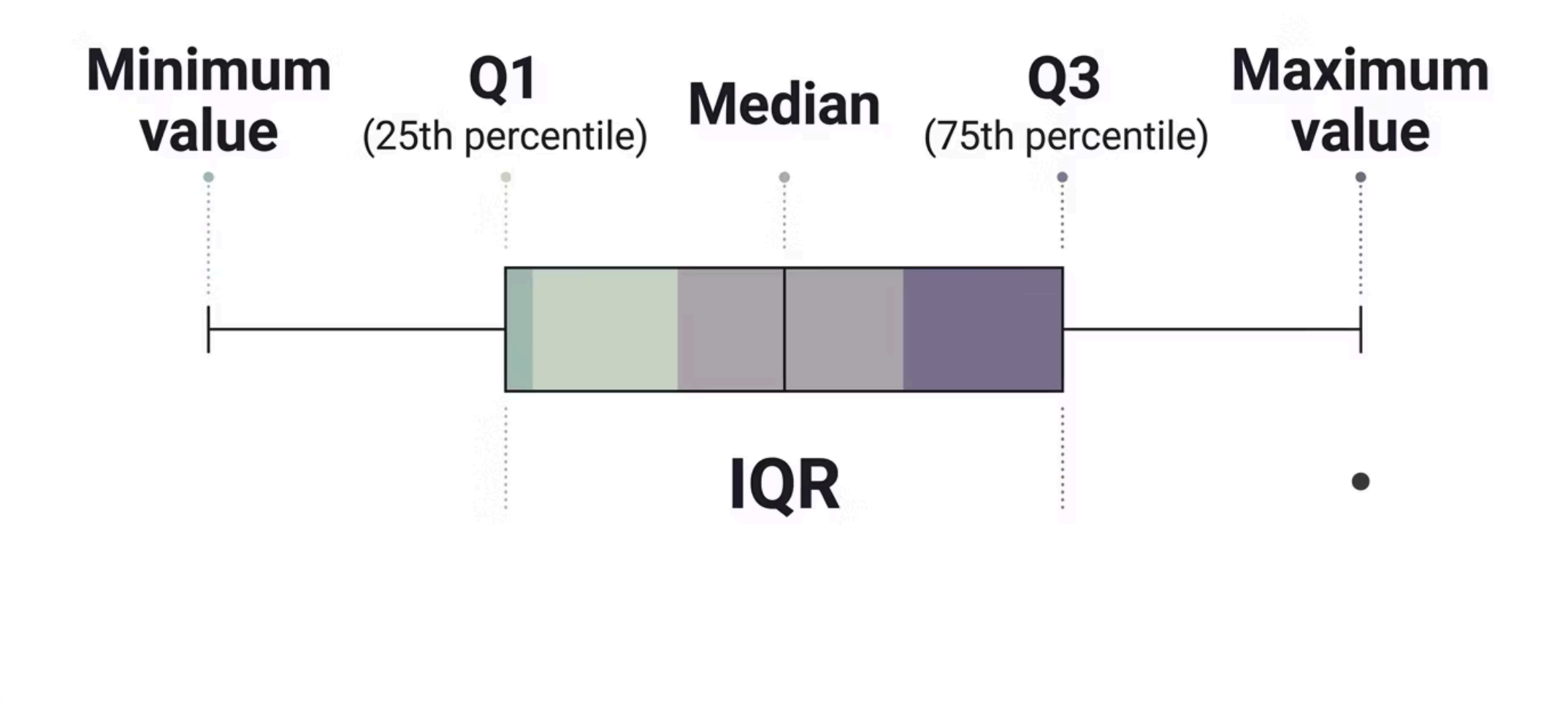
Was the delivery rider lost? Was there an accident? Or is this a data entry error — someone typed 75 instead of 15?

⚠️ **Important:** An outlier is NOT automatically an error. It may be an error, a rare event, or a critical discovery — like fraud.



Box Plot — The Five-Number Summary in One Picture

A box plot compresses a full dataset into five key landmarks. Think of it as a *compressed report card* that shows the important features of your data at a glance.



The Box
 Middle 50% of data. From Q1 to Q3. Width = IQR.

Median Line
 The middle value. Splits the dataset in half.

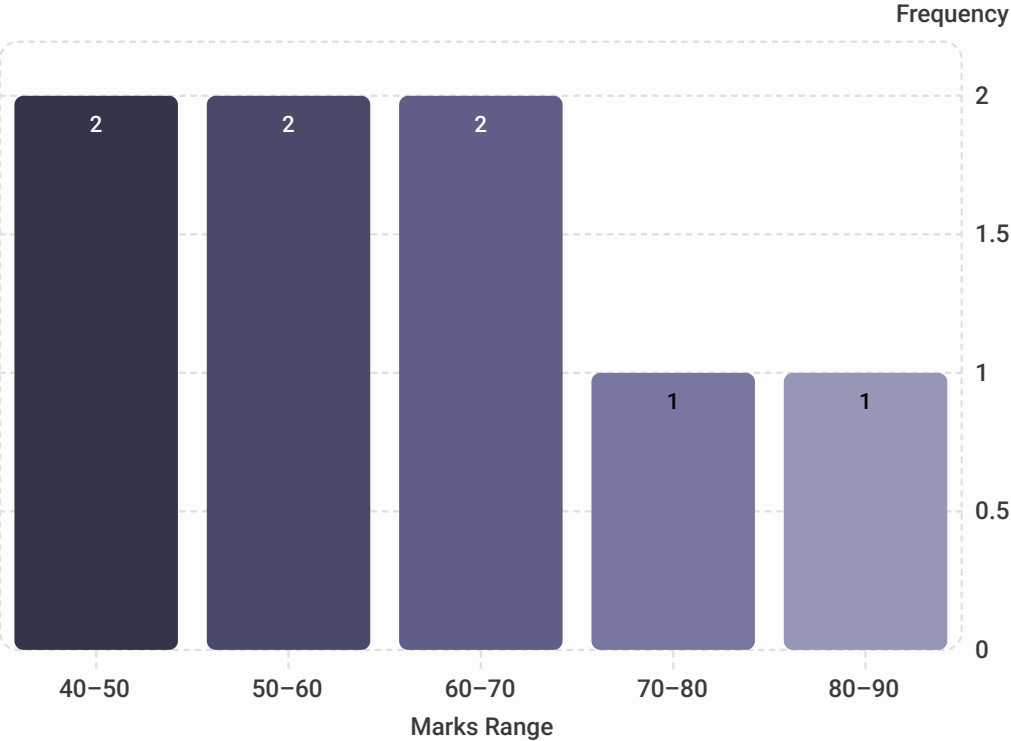
Whiskers
 Extend to the min and max within acceptable range.

Outlier Dots
 Points beyond the whiskers. Worth investigating.

Remember from Lesson 2? $IQR = Q3 - Q1$. Now you can see it visually. The box *is* the IQR.

Histogram vs. Box Plot — Which One Do You Need?

Both show distribution. But they answer *different questions*. Use them together for a complete picture — the histogram tells the story, the box plot gives the summary.



Histogram — Shows the full shape of distribution

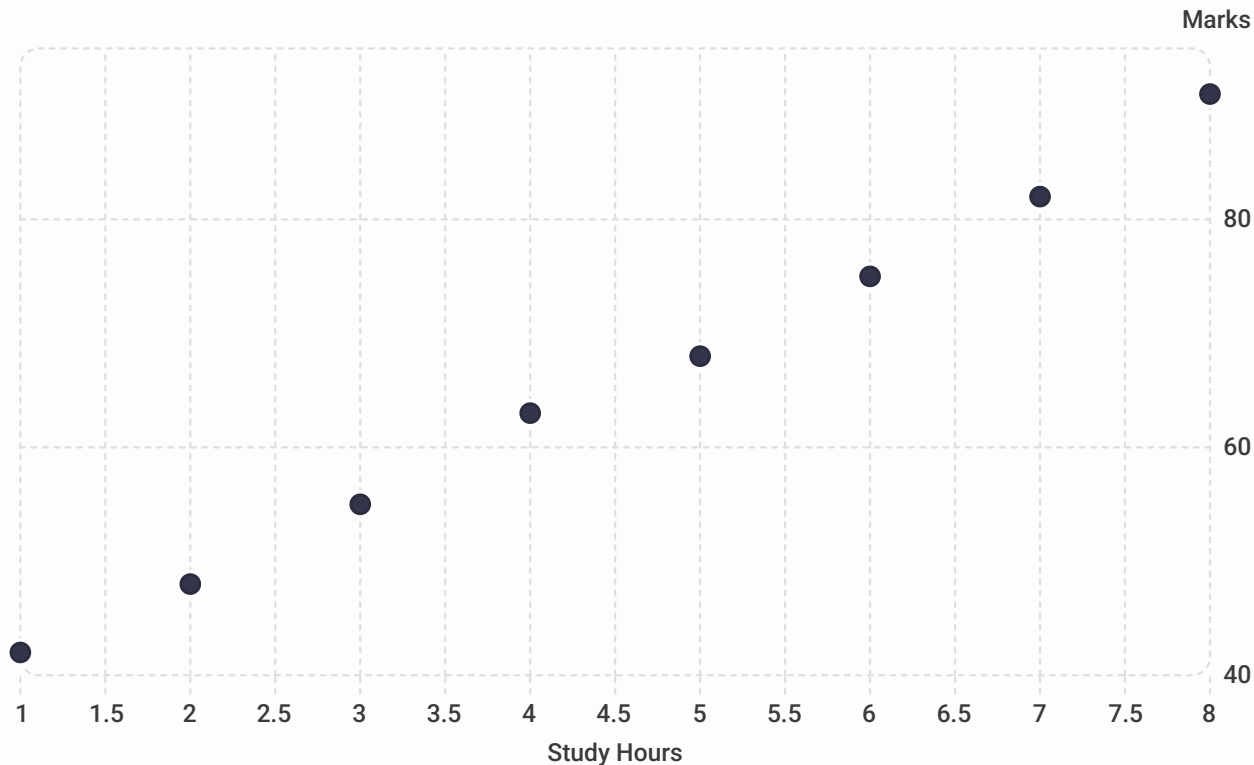
Question	Histogram	Box Plot
Distribution shape	✔ Excellent	⚠ Limited
Outlier detection	⚠ Possible	✔ Excellent
Median visible	✘ Not directly	✔ Yes
Quartiles (Q1, Q3)	✘ No	✔ Yes
Overall spread	✔ Yes	✔ Yes

Box Plot — Gives the five-number quick summary

Scatter Plot — Do Two Variables Move Together?

Each dot is one student. The X-axis shows how many hours they studied. The Y-axis shows the marks they scored. Look at the pattern — what do you notice?

● Marks



What the Scatter Plot Shows

- **X-axis** → Study Hours (input variable)
- **Y-axis** → Marks (outcome variable)
- **Each dot** → One student's data point
- **The trend upward** ↗ → More study = higher marks



Class Question: If study hours increase and marks also increase, what kind of relationship is this?

Positive, Negative & No Relationship

A scatter plot can reveal three types of relationships between two variables. The pattern of the dots tells the story — and the lesson is always more nuanced than it first appears.



Positive Relationship

As X increases, Y also tends to increase.

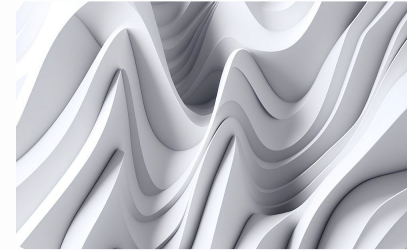
Example: Study Hours vs. Marks. More effort → better result.



Negative Relationship

As X increases, Y tends to decrease. Example:

Absences vs. Marks. More absences → lower marks.



No Clear Relationship

Changes in X show no consistent pattern in Y.

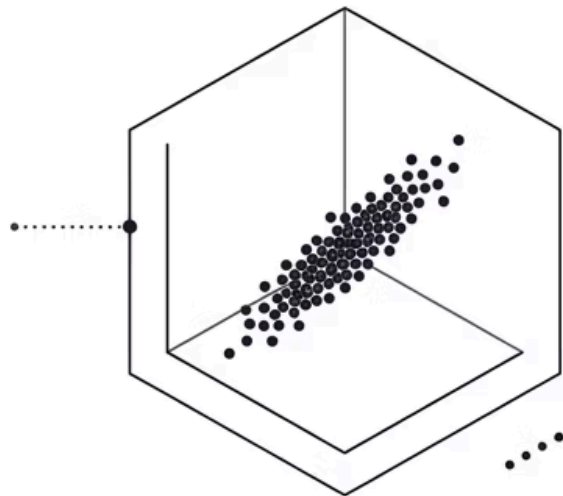
The dots form a random cloud with no direction.

  **CORRELATION ≠ CAUSATION.** Ice cream sales ↑ and sunglasses sales ↑ together — but ice cream doesn't cause people to buy sunglasses. The real cause?  Hot weather. Always ask: is there a hidden third factor?

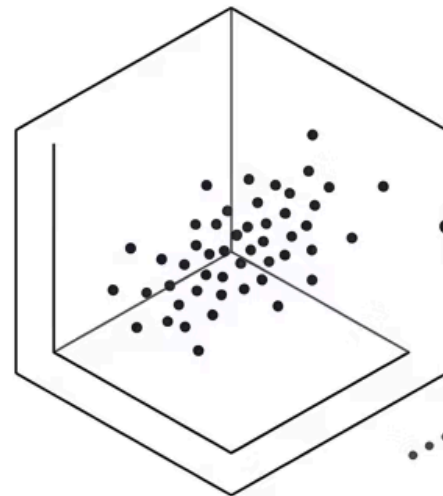
Correlation — Strength & Direction of a Linear Relationship

Correlation is a number that measures how strongly two numerical variables are linearly related. It ranges from **-1 to +1**. The closer to ± 1 , the stronger the relationship.

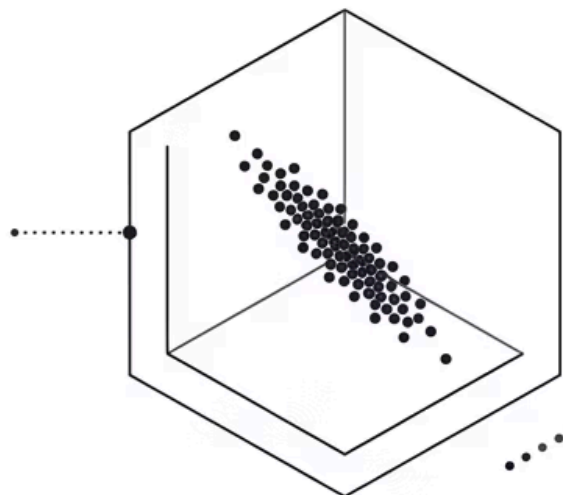
Strong Positive
 $r \approx +0.9$



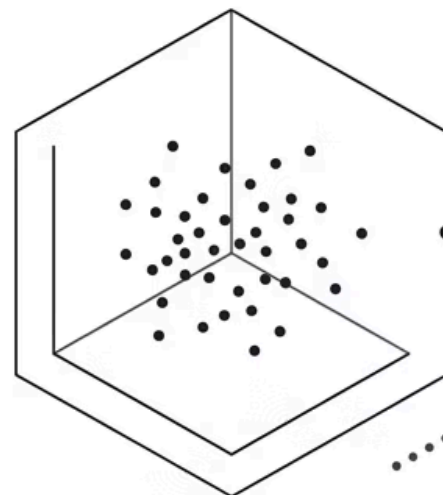
Weak Positive
 $r \approx +0.4$



Strong Negative
 $r \approx -0.9$



No Correlation
 $r \approx 0$



$$r = +1$$

Perfect positive linear relationship

$$r = 0$$

No *linear* association — but a non-linear one may still exist

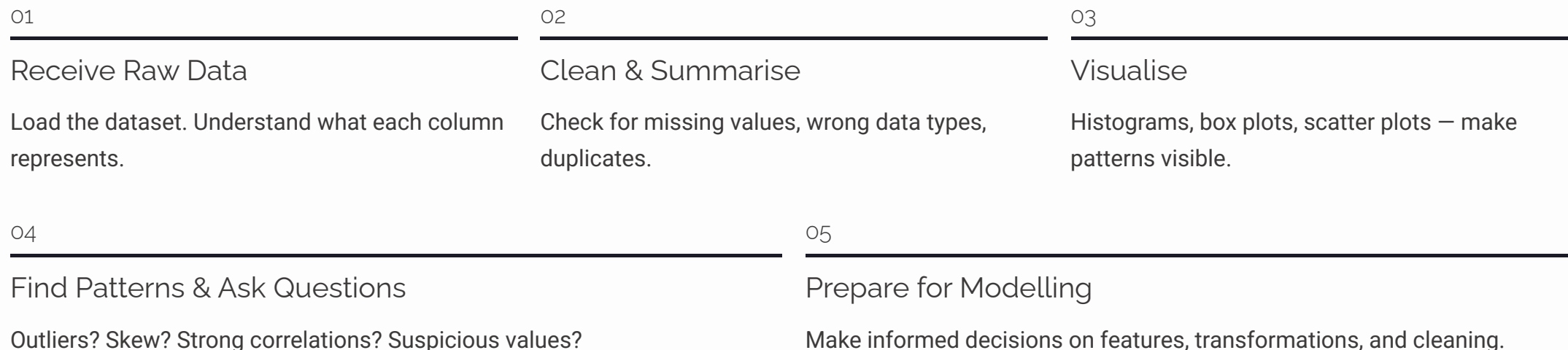
$$r = -1$$

Perfect negative linear relationship

❏ **Important nuance:** $r = 0$ does NOT mean "no relationship whatsoever." It means no *linear* relationship. A curved relationship can still exist.

What Is EDA? — Detective Work Before Modelling

Exploratory Data Analysis (EDA) is the process of examining your dataset *before* building any ML model. A data scientist who skips EDA is like a doctor who prescribes medicine without examining the patient first.



EDA Workflow for a Real ML Project

This is the professional workflow a data scientist follows before writing a single line of model training code. EDA is not optional – it is where most important decisions are made.



📌 **The Golden Rule of ML:** EDA happens *before* blindly training the model. The better your EDA, the better your model's starting point.

AI/ML Case Study — Telecom Customer Churn

A telecom company wants to predict which customers will leave (churn). Before training any model, the data scientist must explore the data. Look at this dataset — what patterns do you notice?

Customer	Data (GB)	Recharge ₹	Calls	Churn?
A	5	199	120	No
B	8	249	150	No
C	15	399	180	No
D	2	199	40	Yes
E	25	599	250	No
F	1	149	20	Yes

What might an ML engineer investigate?

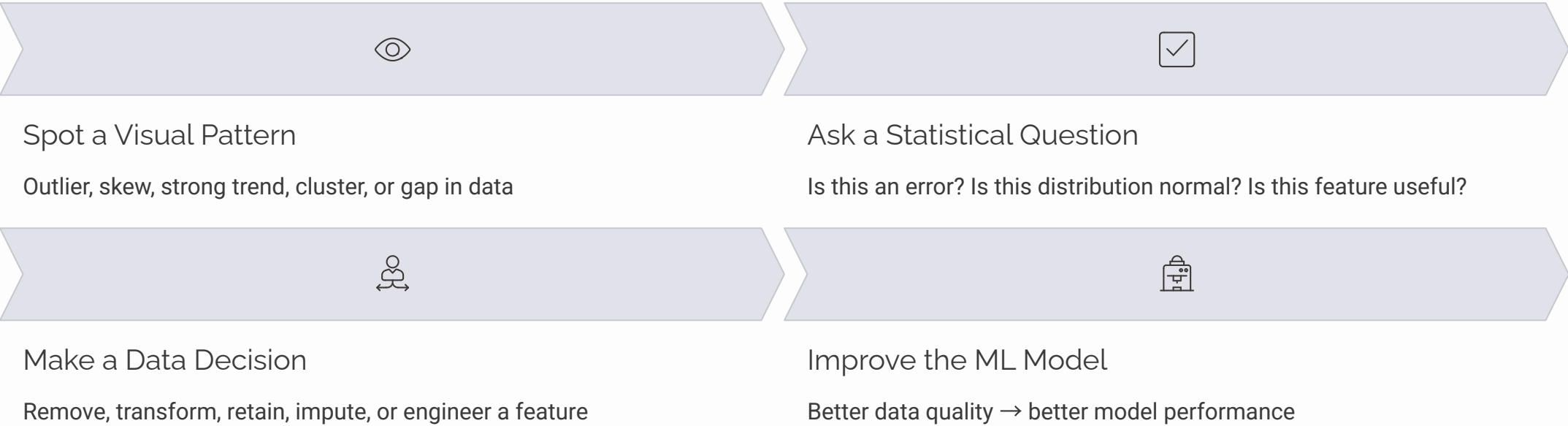
-  **Histogram** of Data Usage → Any extreme values?
-  **Box Plot** of Recharge ₹ → Is there skew? Outliers?
-  **Scatter Plot**: Data GB vs Calls → Do churners cluster?
-  Low usage customers (D, F) are churning — is low engagement a key feature?



Visualizations reveal that **very low usage** may be the strongest early signal of churn — before training a single model.

From Visualization to ML Decision — The Chain of Insight

Visualization does not automatically tell us what to do. It helps us **ask better questions** — and better questions lead to better models.



What You See	What You Ask	What You Do
Outlier detected	Error or genuine event?	Remove / transform / retain
Strong right skew	Should we normalise?	Consider log transformation
Strong correlation between features	Multicollinearity risk?	Feature selection or engineering
Missing values cluster	Missing at random?	Choose imputation strategy

Common Mistakes — Don't Get Tricked by Graphs

Graphs are powerful — but they can also mislead you, especially if you are not careful. Every ML student should know these traps.

✗ Histogram ≠ Bar Chart

Histograms are for *numerical* data in bins. Bar charts are for *categorical* counts. Using the wrong one gives a wrong picture.

✗ Every Outlier Is NOT an Error

Investigate first. An outlier may be fraud, a rare event, or your most important data point.

✗ Correlation ≠ Causation

Two variables moving together does not mean one causes the other. Always look for confounding factors.

✗ Trust the Mean Blindly

If one employee earns ₹2 crore and 9 earn ₹3 lakh, the "average salary" is ₹23 lakh — which represents nobody. Always check the distribution.

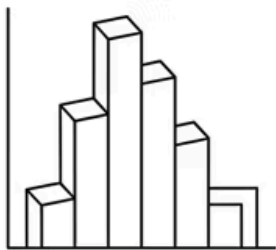
⚠️ **Also watch for:** truncated Y-axes that exaggerate differences, tiny sample sizes that generate false patterns, and ignoring axis units entirely.

Quick Decision Guide — Which Graph Should I Use?

Before picking a visualization, ask yourself: *"What am I trying to discover?"* Use this cheat sheet every time you start exploring a new dataset.

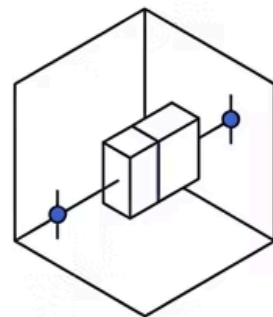
WHAT DO YOU WANT TO DISCOVER?

**DISTRIBUTION
OF ONE
NUMERICAL
VARIABLE**



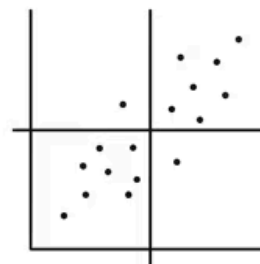
→ Use Histogram

**OUTLIERS
AND SPREAD
QUICKLY**



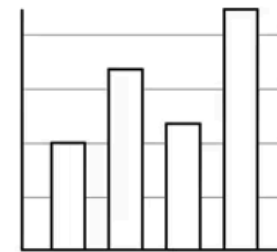
→ Use Box Plot

**RELATIONSHIP
BETWEEN TWO
NUMERICAL
VARIABLES**



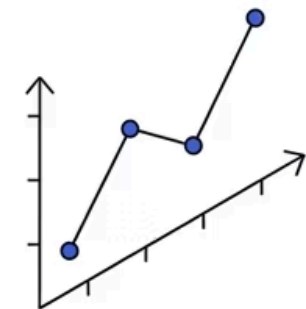
→ Use Scatter Plot

**COMPARE
CATEGORICAL
GROUPS**



→ Use Bar Chart

**CHANGE
OVER TIME**



→ Use Line Chart

 Histogram

One numerical variable →
distribution shape

 Box Plot

Distribution + outliers → quick
five-number summary

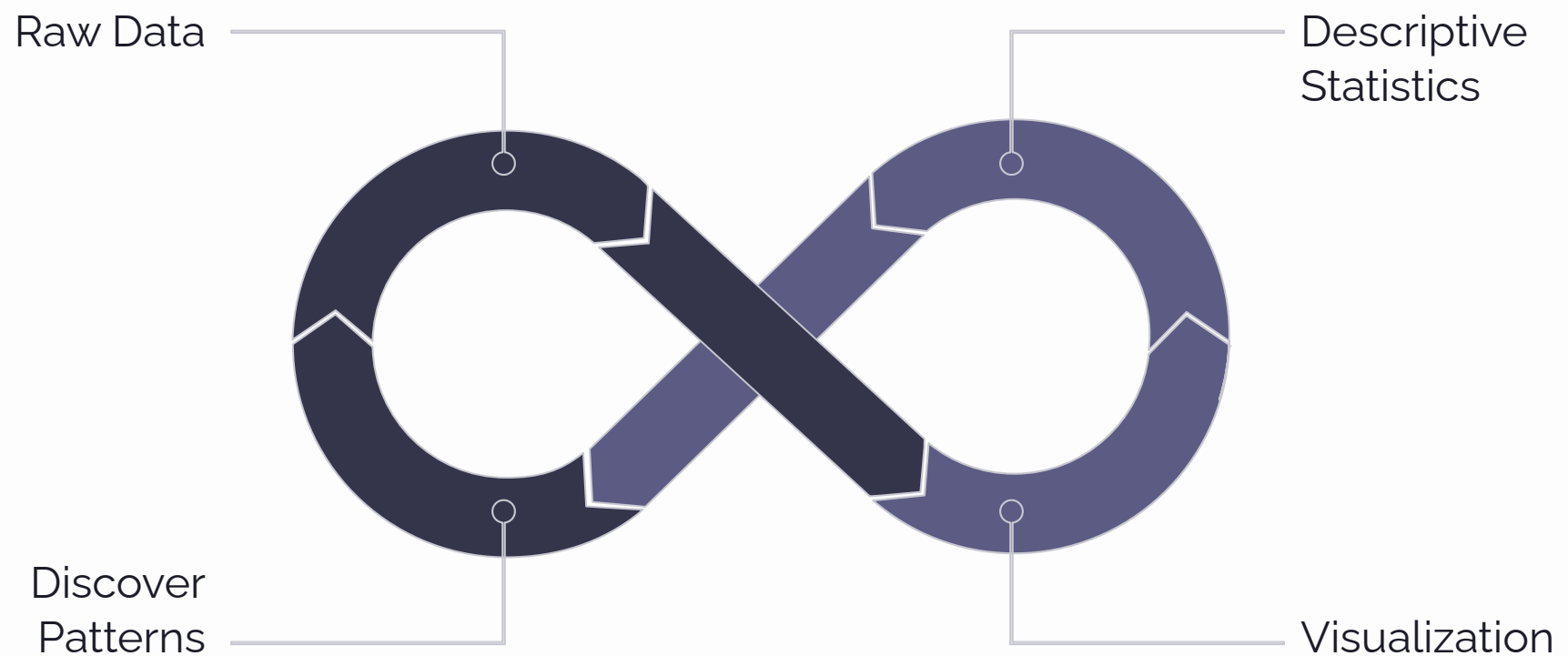
 Scatter Plot

Two numerical variables →
relationship & correlation

 Bar Chart

Categorical counts →
comparing groups

Don't Just Look at Data — Question It.



- 1** What does the distribution look like?
Symmetric, skewed, bimodal? Use a histogram to find out.
- 2** Are there unusual observations?
Investigate every outlier — never assume it is an error.
- 3** How spread out is the data?
IQR and box plots reveal spread better than the mean alone.
- 4** Are variables related?
Scatter plots reveal association — but correlation $\neq$ causation.
- 5** What must we investigate before modelling?
Missing data, skew, outliers, and feature relationships all matter.

"A good data scientist doesn't just calculate numbers. A good data scientist learns to SEE what the numbers are saying."